

Conditional probability language: if-then, given, of and knowing that

use the language of “if then”, “given”, “of”, “knowing that” to describe and interpret situations involving conditional probability.

Learning intention

- use the language of “if then”, “given”, “of”, “knowing that” to describe and interpret situations involving conditional probability.

Teach and model

- using arrays and tree diagrams to represent, interpret and compare probabilities of dependent and independent events.
- using two-way tables and Venn diagrams to understand conditional statements using the language of “if then”, “given”, “of”.
- investigating how conditional probability is used in natural language processing tasks like text or image generation, language translation, data augmentation and recommendation systems.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.